

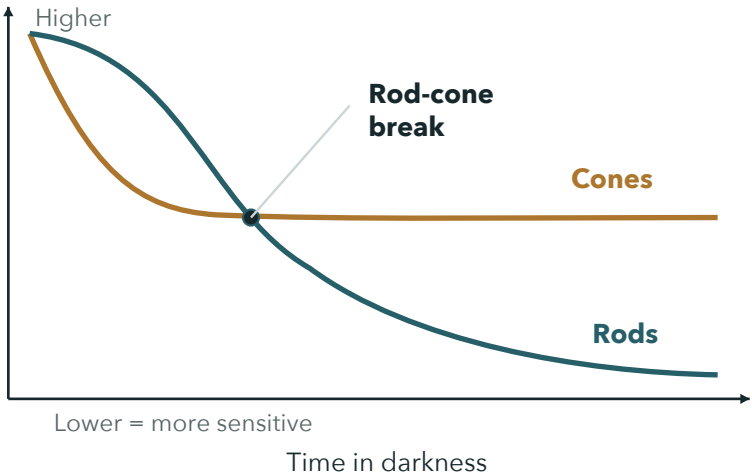
Dark adaptation

Rod recovery, cone speed and the sensitivity-resolution trade-off.

Read the rod-cone break

A | Threshold after bright adaptation

Light threshold



Lower threshold means greater sensitivity. A peripheral test can reveal both branches; a tiny central foveolar test may not.

Which system sets the threshold?

Setting	Main contribution	Useful consequence
Photopic	Cones	Fine detail and colour remain useful
Mesopic	Rods plus cones	Relative brightness can change as rod weighting rises
Scotopic	Rods	High sensitivity, poor colour and fine detail

Fast check

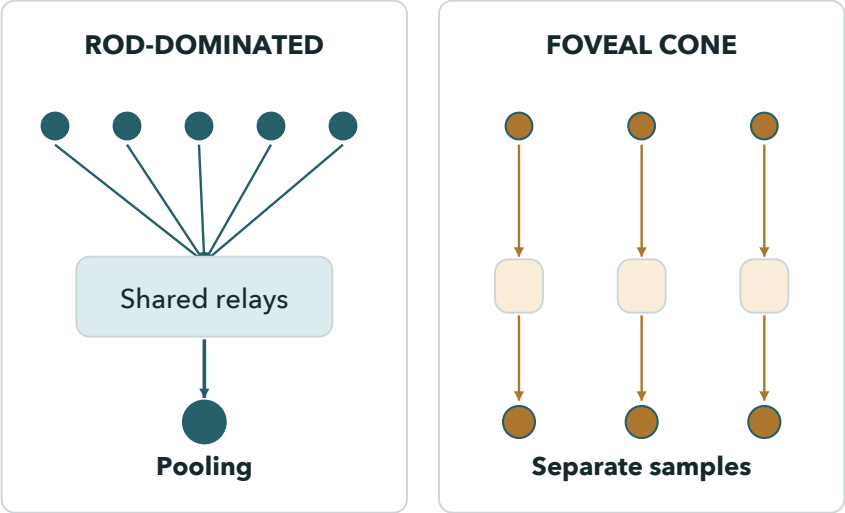
The first rapid fall usually reflects cone recovery; the later rod branch reaches a lower threshold. The crossover is the rod-cone break, not a sudden change in pupil size.

Figure reference

A: Webvision dark adaptation. Original schematic; no fixed timing scale.

Sensitivity costs spatial detail

B | Pooling versus separate sampling



Top circles are receptors; bottom circles are output samples. Rod circuitry includes bipolar and amacrine relays.

Mechanism to observation

Mechanism	Expected observation
Many rod signals converge	Weak light can be detected; nearby points can merge
Low foveal cone convergence	Finer spatial sampling; more light needed
Bright background saturates rods	Cones provide useful increment signalling
Vitamin A limits pigment regeneration	Dim-light difficulty despite normal pupil dilation

Twilight clue

The Purkinje shift makes blue-green relatively brighter than red as rod contribution rises. Rods signal brightness without normal trichromatic colour discrimination.

Figure reference

B: rod/cone distribution and photoreceptors; Purkinje study. Original schematic.